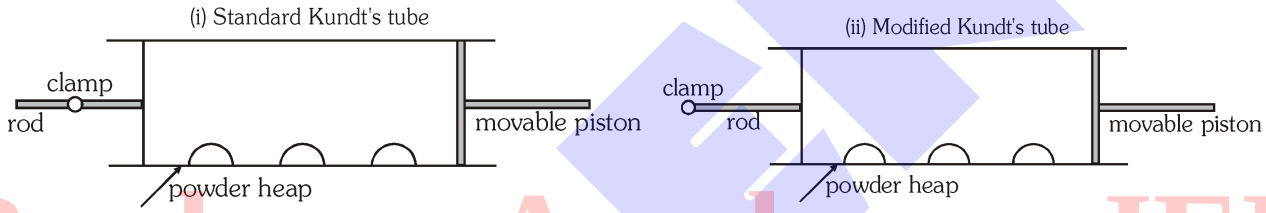
SECTION-I

Single Correct Answer Type 9 Q. [3 M (–1)]

1. The speed of sound in neon (Ne) at a certain temperature is V ms^{-1}. The speed of sound in hydrogen (H_2) at the same temperature will be (assume both gases to be ideal)

(A) $V\sqrt{\frac{42}{5}}$ ms^{-1} (B) $V\sqrt{\frac{5}{42}}$ ms^{-1} (C) $V\sqrt{5}$ ms^{-1} (D) $\frac{V}{\sqrt{5}}$ ms^{-1}

2. In the Kundts tube experiment (shown in fig. (i)), the rod is clamped at the end instead of clamping it at the center as shown in fig. (ii). It is known that speed of sound in air is 330 m/s, powder piles up at successive distance of 0.6 m and length of rod used is 1m, speed of sound in rod is



(A) 550 m/s (B) 1100 m/s (C) 1200 m/s (D) 600 m/s

3. Two identical wires under the same tension have a fundamental frequency of 500 Hz. The fractional increase in the tension of one wire will give 5 beats per second, is :-

(A) 0.01 (B) 0.02 (C) 0.03 (D) 0.04

4. A sonometer has two strings stretched side by side parallel to each other. The two strings are made of the same metal and have uniform cross-sections of radii r_1 and r_2 respectively with $r_1 : r_2 :: 2 : 3$. The two strings are each of length 1m and are loaded by identical weights. The first string is made to oscillate with six loops in its length. How many loops will be seen on the second string when it resonates with the first string?

(A) 2 (B) 3 (C) 9 (D) 6

5. **Statement 1 :** When a closed organ pipe vibrates, the pressure of the gas at the closed end remains constant.

Statement–2 : In a stationary–wave system, displacement nodes are pressure antinodes, and displacement antinodes are pressure nodes.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

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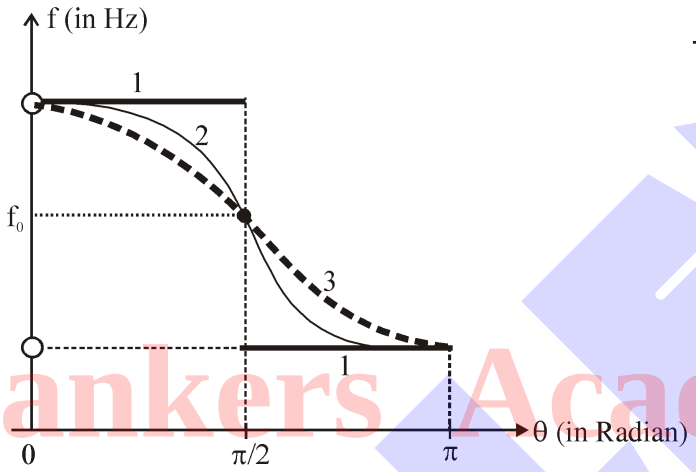
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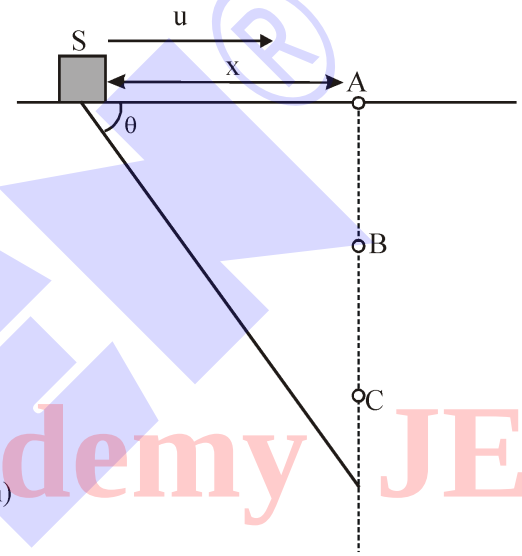
6. A bird is singing on a tree and a man is hearing at a distance 'r' from the bird. Calculate the displacement of the man towards the bird so that the loudness heard by man increases by 20 dB.

[Assume that the motion of man is along the line joining the bird and the man]

(A) $\frac{9r}{10}$ (B) $\frac{r}{10}$ (C) $\frac{3r}{5}$ (D) $\frac{4r}{5}$

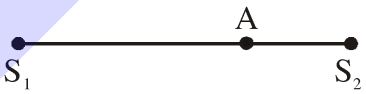
7. A source is moving with a constant speed u on a straight line, emitting a sound of frequency f_0. There are three observers A, B and C. A on track, B at a perpendicular distance of d from the track and C at a perpendicular distance of 2d form the track as shown in the figure. The variation of the observed frequency with respect to the position x,





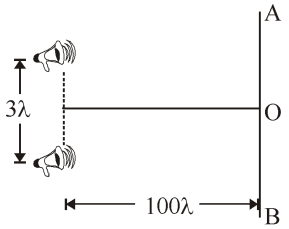
(A) A – 3, B – 2, C – 1 (B) A – 2, B – 3, C– 1 (C) A – 1, B – 2, C – 3 (D) A – 1, B – 3, C – 2

8. If $y_1 = 5$ (mm) $\sin\pi t$ is equation of oscillation of source S_1 and $y_2 = 5$ (mm) $\sin(\pi t + \pi/6)$ be that of S_2 and it takes 1 sec and ½ sec for the transverse waves to reach point A from sources S_1 and S_2 respectively then the resulting amplitude at point A, is



(A) $5\sqrt{2+\sqrt{3}}$ mm (B) $5\sqrt{3}$ mm (C) 5 mm (D) $5\sqrt{2}$ mm

9. Two loudspeakers are emitting sound waves of wavelength λ with an initial phase difference of $\frac{\pi}{2}$. At what minimum distance from O on line AB will one hear a maxima ?



(A) 25 λ (B) $\frac{100\lambda}{\sqrt{15}}$

(C) $\frac{25\lambda}{3}$ (D) 50 λ

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Multiple Correct Answer Type 8 Q. [4 M (–1)]

10. A small source of sound operates at 1kHz,24πW emitting sound uniformly in all directions.The speed of sound is 360m/s in air and density of air is 1.2kg/m^3.

(A) The intensity of sound at a distance 10m from the source is 0.06W/m^2

(B) The intensity of sound at a distance 10m from the source is 0.24W/m^2

(C) The pressure amplitude at distance 10m is 7.2 Pa

(D) The pressure amplitude at distance 10m is 28.8 Pa

11. In a resonance column experiment, the length of air column at 1st resonance is measured to be 13.1 cm with a possible indeterminate error of 0.1 cm. With the same scale, the 2nd resonance occurs at 39.2 cm. The possible wavelength of the sound can be :

(A) 52.2 cm (B) 52.4 cm (C) 52.5 cm (D) 51. 9 cm

12. A narrow steel rod of length 'L' is rigidly clamped at its mid-point and transverse standing waves of frequency 'f' are set up in it. The speed of transverse waves in the rod is 'c'.Then,

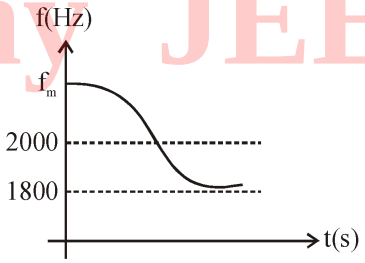
(A) The free ends of the rod must be antinodes.

(B) The fundamental frequency of the rod is c/(L)

(C) The second overtone frequency of the rod is 5c/(2L)

(D) 'f' can be any integral multiple of the fundamental frequency.

13. A stationary observer receives a sound of frequency f_0 = 2000 Hz when the source is at rest. When the source starts moving with a constant velocity starting from a large distance, the apparent frequency f varies with time as shown in figure. Speed of sound = 300 m/s. Choose the **CORRECT** alternative (s):-



(A) Speed of source is 66.7 m/s.

(B) f_m shown in figure can be 2500 Hz.

(C) Speed of source is 33.33 m/s.

(D) f_m shown in figure cannot be greater than 2250 Hz.

14. Choose the incorrect one

(A) When an ultrasonic wave travels from air into water, it bends towards the normal to the air water interface

(B) Any function of the form y(x, t) = f(vt + x) represents a travelling wave

(C) The velocity, wavelength and frequency of wave undergo change when it is reflected from a surface

(D) None of the above

15. A person blows into open-end of a long pipe. As a result, a high-pressure pulse of air travels down the pipe. When this pulse reaches the other end of the pipe. **[JEE 2012]**

(A) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is open

(B) a low -pressure pulse starts travelling up the pipe, if the other end of the pipe is open

(C) a low pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

(D) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

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16. An open organ pipe of length 24 cm and an organ pipe closed at one end of length 38 cm have the same diameter. It is found that the 1st overtone of closed pipe is in unison with the fundamental of the open pipe. The velocity of sound in air is 340 ms^{-1} . Choose the CORRECT statement(s)

(A) The end-correction at each open end is 1cm

(B) the fundamental frequency of the open pipe is nearly 654 Hz

(C) The frequency of 2nd harmonic of open pipe will be twice its fundamental frequency

(D) The frequency of 2nd harmonic of open pipe will be less than twice its fundamental frequency.

17. Mark the **CORRECT** statement (s) :-

(A) When two waves of different frequency f_1 and f_2 are superimposed at a point, the variation of intensity takes place with frequency $f_1 - f_2$

(B) When an observer is moving, relative speed of the sound waves with respect to the observer is changed, while due to motion of source, the sound wavelength of the wave changes.

(C) When a high pressure compression pulse reaches water surface in a resonance column, it reflects as a low pressure rarefraction pulse due to a phase change of π

(D) When a tuning fork is struck by a hammer a transverse wave sets in the prong while a longitudinal wave sets in the stem (handle) of tuning fork

Linked Comprehension Type (1 Para × 4Q.) (1 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question No. 18 to 21

Imagine that you are in Indian army and you have been given charge of commanding parade on 26th January. The soldiers have done extensive training on march past with a band which strikes 120 times/min. The soldier should strike left foot or right foot at each drum beat so that parade marches at 120 steps/min.

On the day of parade, there are 2 parts. In 1st part, the band marches ahead of the soldiers with same speed as that of soldiers. In this march, you notice that the soldiers in the first row strike left foot but the soldiers in last row strike right foot at the same time. Unfortunately, the general also notices this.

In the 2nd part, you hope & pray that everything goes well. In this part, the band is stationary but the soldiers advance towards the band at same constant speed. Here again, you notice that the soldiers in the parade are doing 121 steps/min. As before, the general notices this.

A court martial is due for you. You frantically try to look for an explanation. One of your friends reminds you of your old physics teacher at Allen. You turn to him for solace. He suggests two things for defence.

(a) Sound takes time to reach from one place to another.

(b) doppler's effect

Assume that it was chilly winter & speed of sound = 320 m/sec.

18. Select the correct explanation for mistakes in each part.

(A) part I $\rightarrow$ a, part II $\rightarrow$ b (B) part I $\rightarrow$ a, part II $\rightarrow$ a

(C) part I $\rightarrow$ b, part II $\rightarrow$ a (D) part I $\rightarrow$ b, part II $\rightarrow$ b

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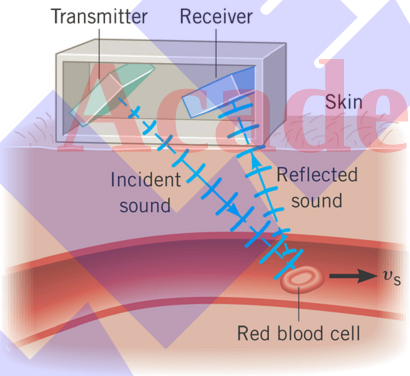
19. What can be the length of the column of soldiers marching past.
(A) 320 m (B) 160 m (C) 80 m (D) 40 m

20. What is the speed of the soldiers.
(A) 1.0 m/s (B) 2.0 m/s (C) 1.5 m/s (D) 2.5 m/s

21. What will be the marching speed of the soldiers when they turn around & march away from the band in part 2 ?
(A) 120 steps/min. (B) 119 steps/min. (C) 121 steps/min. (D) 110 steps/min.

Paragraph for Questions no. 22 to 24

The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.



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22. Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx) speed of the sound wave in the blood?
(A) 1700 m/s (B) 330 m/s (C) 5000 m/s (D) 3000 m/s

23. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region?
(A) 150 Hz (B) 300 Hz (C) 600 Hz (D) 2400 Hz

24. At what extra rate does the heart have to work due to this narrowing down of the artery? Assume the density to be 1.5 gm/cc and the area of the normal artery to be 0.1 cm^2.
(A) 1.125×10^{-4} W (B) 2.5×10^{-4} W (C) 6.25×10^{-5} W (D) 5.625×10^{5} W

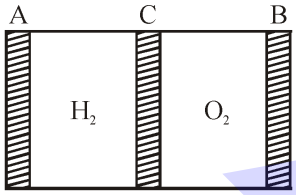
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SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

25. AB is a cylinder of length 1m fitted a thin flexible diaphragm C at the middle and other thin flexible diaphragms A and B at the ends. The portions AC and BC contain hydrogen and oxygen gases respectively. The diaphragms A and B are set into vibrations of same frequency. What is the minimum frequency of these vibration for which diaphragm C is a node ? (Under the conditions of experiment $v_{H_2} = 1100$ m/s, $v_{O_2} = 300$ m/s). **[IIT-1978]**



SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

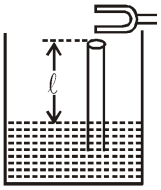
26. A super car is moving with velocity 150 m/sec towards a fixed wall producing a sound of frequency 75 Hz. Speed of sound is 300 m/sec then find the wavelength (in m) of reflected sound.

27. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length ℓ(in cm) of the pipe and radius r (in cm) of the pipe are $\ell = 94 \pm 0.1$, $r = 5 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment is given by $\alpha^2\%$. Find the value of 10α.

28. A plane longitudinal wave having angular frequency $\omega = 500$ rad/s is travelling in positive X direction in a medium of density $\rho = 1$ kg/m^3 and bulk modulus 4×10^4 N/m^2 . The loudness at a point in the medium is observed to be 20dB. The pressure amplitude of the pressure wave is given has $\alpha \times 10^{-4}$ N/m^2. Find the value of α.

29. A ball of radius R = 100 cm is located in the way of propagation of a plane sound wave. The sonic wavelength is $\lambda = 2$ cm, the frequency is $\nu = 100\pi$ Hz, the pressure oscillation amplitude in air is $(\Delta p)_m = 4$ Pa. Find the mean energy flow rate (in watt), averaged over an oscillation period, reaching the surface of the ball. Take density of air 1 kg/m^3.

30. A tuning fork of known frequency is held at the open end of a long tube which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates when air columns have length $\ell_1 = 50 \pm 0.5$ cm and $\ell_2 = 150 \pm 0.1$ cm. If the speed of sound in air is $[330 \pm \alpha\,(0.99)]$ m/s then the value of α is _____



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Subjective Type **1 Q. [4 M (0)]**

31. A point isotropic sonic source of sound power 1 milli watts emits sound of frequency 170 Hz in all directions.

(a) Find the distance of a point from the source where the loudness level is 60 dB. [Take $v_s = 340$ m/s]

(b) It is known that at some moment t = 0, the displacement of air particles at a certain point 4 m away from the source is A. The amplitude of oscillation at this point is known to be 2A. Find the displacement of air particles in terms of A, at a point 55 m away from the source at the moment t = 0. Consider displacement of particles to be positive if it is away from the source.

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GR # SOUND WAVE **ANSWER KEY**

SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (B)** **3. Ans. (B)** **4. Ans. (C)** **5. Ans. (D)**
6. Ans. (A) **7. Ans. (C)** **8. Ans. (C)** **9. Ans. (C)**

Multiple Correct Answer Type **8 Q. [4 M (–1)]**

10. Ans. (A, C) **11. Ans. (A, B, C, D)** **12. Ans. (A,C)** **13. Ans. (C,D)**
14. Ans. (A,C) **15. Ans. (B,D)** **16. Ans. (A, B, C)** **17. Ans. (A,B, D)**

Linked Comprehension Type **(1 Para × 4Q.)** **(1 Para × 3Q.)** **[3 M (-1)]**
(Single Correct Answer Type)

18. Ans. (A) **19. Ans. (B)** **20. Ans. (D)** **21. Ans. (B)** **22. Ans. (A)**
23. Ans. (D) **24. Ans. (A)**

SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**
(upto second decimal place)

25. Ans. 1650 Hz

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

26. Ans. 2 **27. Ans. 4** **28. Ans. 2** **29. Ans. 4** **30. Ans. 2**

Subjective Type **1 Q. [4 M (0)]**

31. Ans. (a) $r = \sqrt{\frac{250}{\pi}}$ m, (b) $-\frac{4A}{55}$

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